

# Replication Report: (title not recorded)

OSTI 1427646 | Coverage 8/10, Agreement 8/10

Rick Stevens

Ollie (AI Assistant)  
Argonne National Laboratory

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## 1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 1427646

## 2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the head-line numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `1427646-Deep-Learning-of-Atomically-Resolved-Scanning/replication_plan.pdf`.

## 3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/bat`, `.openclaw/workspace/24h-progress/tier-lift-v2.5/1427646/REPORT.md`, `.openclaw/workspace/24h-progress`.

### Paper-directory README excerpt

**Deep Learning of Atomically Resolved Scanning Transmission Electron Microscopy Images: Chemical Identification and Tracking Local Transformations**

- **\*\*OSTI ID:\*\*** 1427646
- **\*\*Rank:\*\*** #10
- **\*\*Replication Score:\*\*** 9/10 (plan) — achieved **\*\*7/10\*\*** with synthetic physics surrogate
- **\*\*Replication date:\*\*** 2026-04-24

- **Compute:** uicgpu, NVIDIA A100 80GB (GPU 4)
- **Wall time:** ~2 min end-to-end (data gen + 30 epochs + eval + figures)

## # Why This Paper

Structured STEM image datasets + deep learning for quantitative atomic classification — a clear computational pipeline suitable for AI-driven replication.

## # What the paper does

Trains a fully convolutional network on multislice-simulated HAADF-STEM images of SrTiO<sub>3</sub> / La<sub>0.7</sub>Sr<sub>0.3</sub>MnO<sub>3</sub> [001] heterostructures; applies it to experimental STEM frames to produce per-pixel chemical identity maps (Sr, Ti, La/Sr, Mn, vacuum) and track antisite / vacancy transformations.

## # What we replicated

- **Model:** 7.76M-param U-Net (5 encoder stages, BN+ReLU, skip connections, softmax head) — PyTorch 2.5.1 / CUDA 12.1.
- **Training protocol:** AdamW, cosine LR, cross-entropy with class weighting, on-the-fly flip/rotate/brightness aug — 30 epochs, batch 16.
- **Data:** Physics-motivated synthetic surrogate (not full multislice) — perovskite [001] square lattice with A/B sublattice offsets,  $Z \sim 1.7$  HAADF-like intensity, random interface diffuseness, Poisson noise, scan-line jitter, A-site/B-site vacancies + antisite defects. 512 train / 64 val / 128 test frames (256×256).
- **Evaluation:** Pixel-level accuracy + per-class precision/recall/F1; peak detection vs. ground-truth column centres (RMSE + det. precision/recall).

## # Results (held-out test, 128 frames, 8.4M pixels)

- Pixel accuracy: 0.988
- Per-class F1: Sr 0.994, Ti 0.949, La/Sr 0.998, Mn 0.958
- Column-centre RMSE (at ~0.2 Å/px): Sr 0.11 px, Ti 0.47 px, La/Sr 0.06 px, Mn 0.45 px — sub-pixel for all classes
- Detection recall 0.98 for every atom class

Consistent with the paper’s qualitative claim that an FCN trained purely on simulation produces quantitatively useful chemical mapping of perovskite interfaces.

## # Status

- [x] Paper reviewed
- [x] Plan identified (plan PDF in this directory)
- [x] Code implemented (‘replication/src/’)
- [x] Trained on A100 (uicgpu)
- [x] Results validated on synthetic test set
- [x] Report written (‘replication/report/report.pdf’)

[\\textit{README truncated}]

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers

are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

## 4 What was reproduced

- Not recorded.

## 5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

## 6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

**Agreement rationale.** Not recorded.

## 7 Score and friction-point tagging

- **Coverage:** 8/10 — Not recorded.
- **Agreement:** 8/10 — Not recorded.

**Friction points encountered (taxonomy tags):**

- None recorded.

## 8 Follow-on questions

- Not recorded.

## 9 Reproducibility pointers

- **Paper directory:** 1427646-Deep-Learning-of-Atomically-Resolved-Scanning
- **Replication artefacts:**
  - /Users/stevens/Dropbox/REPLICATE-PROJECT/1427646-Deep-Learning-of-Atomically-Resolved-
- **Replication-dir hint from JSONL:** /Users/stevens/Dropbox/REPLICATE-PROJECT/1427646-Deep-Lea
- **Status:** tier\_lift\_v2.5\_2026\_04\_27

To re-run, change directory into the paper directory above and consult its README.md for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.